# ReliableML Method Note: Distribution Shift and Drift

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**Abstract**

The drift monitor compares incoming feature batches with the training reference. It supplies unlabeled input-distribution evidence and, separately, a labeled feature–target divergence diagnostic. Neither is a direct test of prediction correctness.

## 1 Reference and incoming data

`DriftMonitor.fit(X_train)` records profiles for each named feature. Incoming batches must preserve the fitted feature names, order, and compatible representation. Numerical profiles use reference quantile bins; categorical profiles retain observed categories and an unseen-category bucket. Missing, unseen, underflow, and overflow values are tracked separately where applicable.

## 2 Statistics

For numeric and categorical features, the report computes the population stability index (PSI) and smoothed Kullback–Leibler divergence on the fitted bins. Numeric features additionally receive a two-sample Kolmogorov–Smirnov statistic and one-dimensional Wasserstein distance when both batches contain finite observations. A library-defined Drift Score Index measures changes in mean, standard deviation, skewness, and excess kurtosis; scale changes are normalized by the reference standard deviation. Categorical changes use a Pearson chi-square homogeneity test over shared and unseen categories.

The report can include bootstrap intervals for applicable metrics, drift flags, and aggregate summaries. Thresholds are user-configurable and should be validated against representative in-domain and shifted data. The PSI-derived `similarity_score` is a descriptive transform, not a probability, calibrated confidence, or guarantee of model validity.

## 3 Feature–target divergence

With labels, the concept-shift diagnostic compares a univariate approximation of feature–target associations between a reference and incoming batch. This does not identify a causal mechanism and can miss changes in interactions or conditional relationships. It is a separate labeled analysis, not part of the unlabeled inference-time drift score.

## 4 Operations and limitations

Numeric KS and Wasserstein calculations exclude missing and nonnumeric observations. Categorical chi-square approximations can be unreliable with small expected counts. Bootstrap intervals

assume the implemented row resampling scheme; dependent observations may require block or cluster bootstrap. Distribution changes can be benign, and stable marginals can hide conditional drift. Evaluate performance on labeled, representative holdout data before operational decisions.

## 5 Interface

```
monitor = DriftMonitor().fit(X_train)
report = monitor.assess(X_incoming)
print(report.to_dict())
```

Through `ReliablePipeline`, the same fitted reference profile is used when predicting and reporting on unseen feature batches.

## 6 Further reading

## References


[1] J. Gama et al. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4), 2014. https://doi.org/10.1145/2523813.

[2] ReliableML implementation and documented formulas: https://github.com/Jabualani/ML-Reliabilty.